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A World of Vibrations

Quantum Field Theory

The phrase “action at a distance,” usually modified by Einstein’s adjective “spooky,” is often invoked in discussions of quantum entanglement and the EPR puzzle. But the idea is much older than that—it goes back at least to Isaac Newton and his theory of gravity.

If Newton had done nothing more than put together the basic structure of classical mechanics, he would be a leading candidate for the greatest physicist of all time. What clinches his claim to the crown is that he did much more than that, including little things like inventing calculus. Still, when most people see a picture of Newton in his magnificent wig, they think of his theory of gravity.

Newtonian gravity can be summed up in the famous inverse-square law: the gravitational force between two objects is proportional to the mass of each of them, and inversely proportional to the square of the distance between them. So if you moved the moon to be twice as far away from the Earth, the gravitational force between them would be only one-fourth as large. Using this simple rule, Newton was able to show that planets would naturally move in ellipses around the sun, confirming the empirical relationship that had been posited by Johannes Kepler years before.

But Newton was never really satisfied with his own theory, precisely because it featured action at a distance. The force between two objects depends on where each of them is located, and when an object moves, the direction of its gravitational pull changes instantaneously all throughout the universe. There was nothing in between that would mediate such a change;